

E9 205 Machine Learning for Signal Processing

Neural Networks

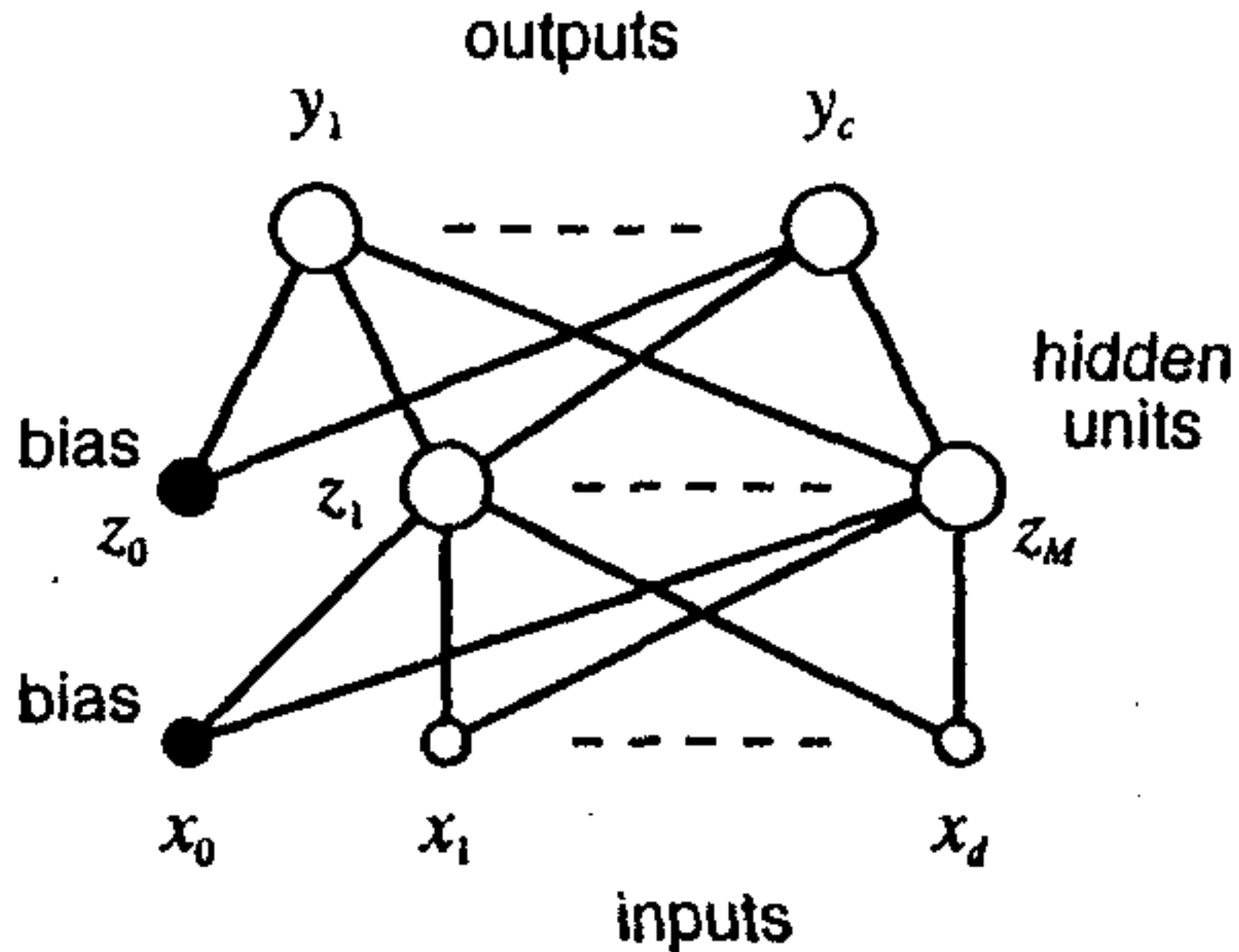
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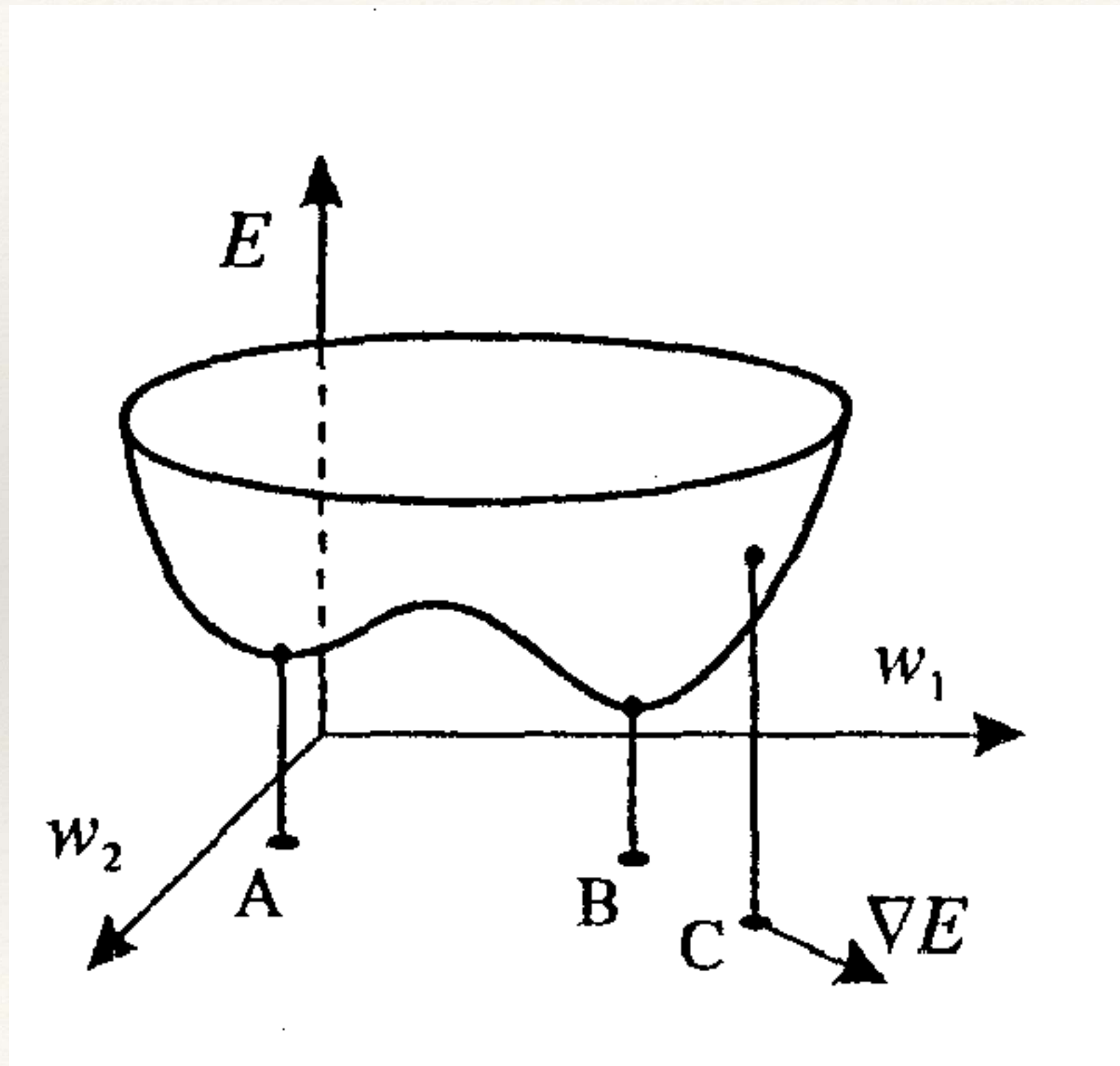
Neural Networks - Recap



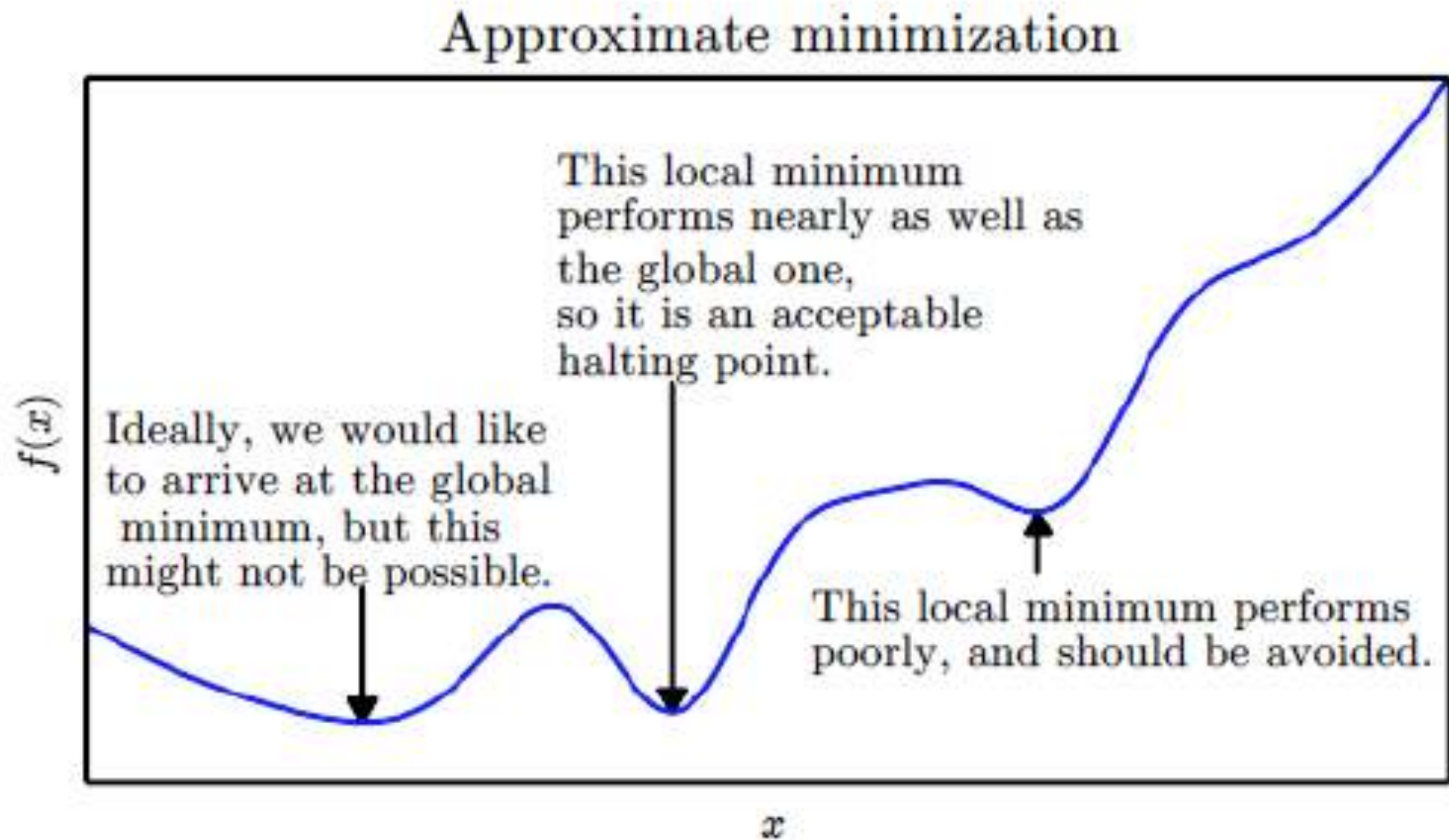
Neural Networks - Recap

- ❖ Need for multiple hidden layers
- ❖ Activation function at hidden layers.
- ❖ Computation of gradient using back propagation.
- ❖ Error function and output layer activation
 - ❖ Linear activation with MSE loss
 - ❖ Cross entropy loss function
- ❖ Neural networks estimate posterior probabilities
 - ❖ Shown for MSE loss function.

Typical Error Surface

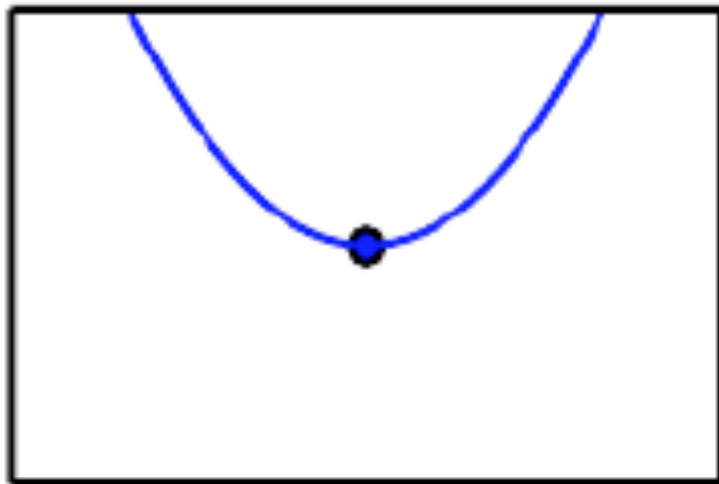


Error Surface in 1-D

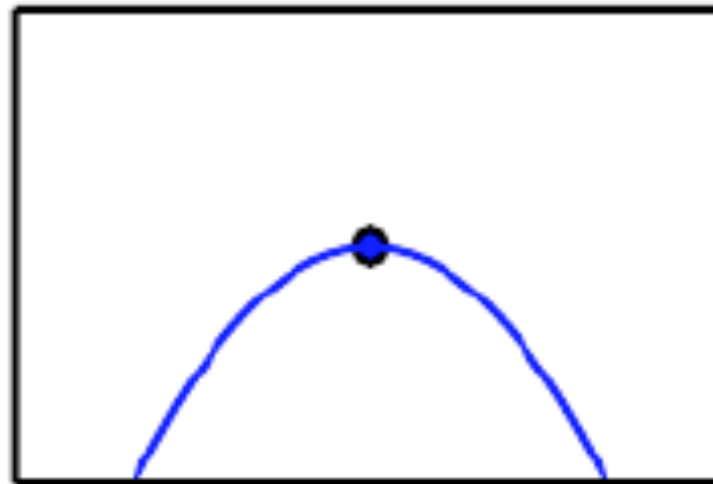


Saddle Points

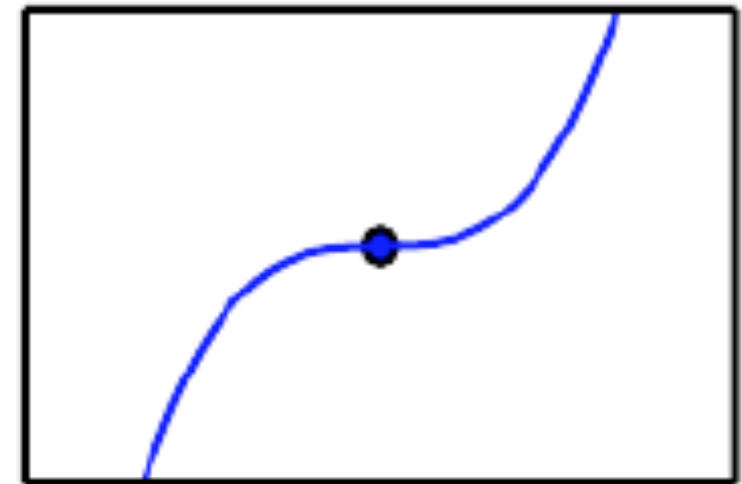
Minimum



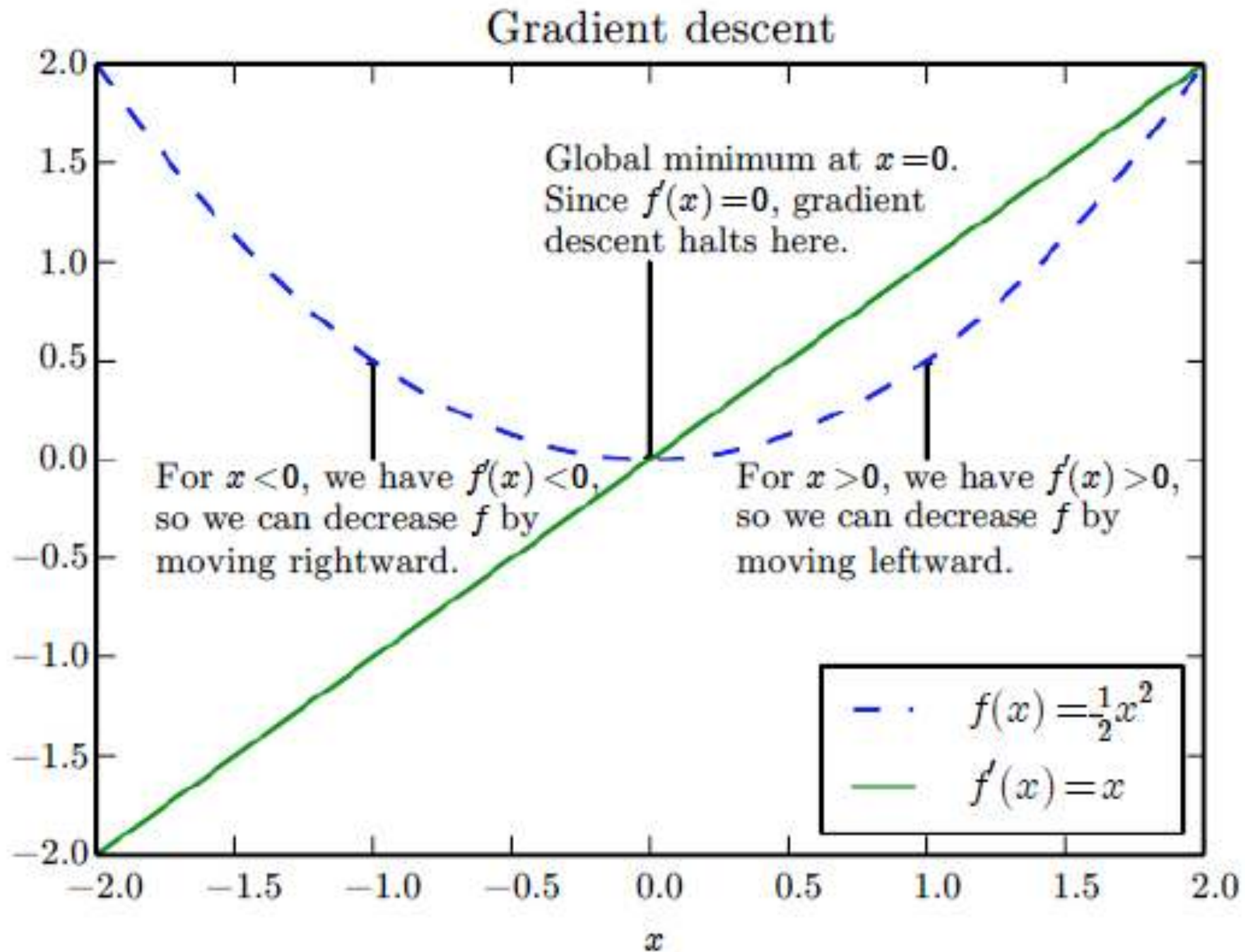
Maximum



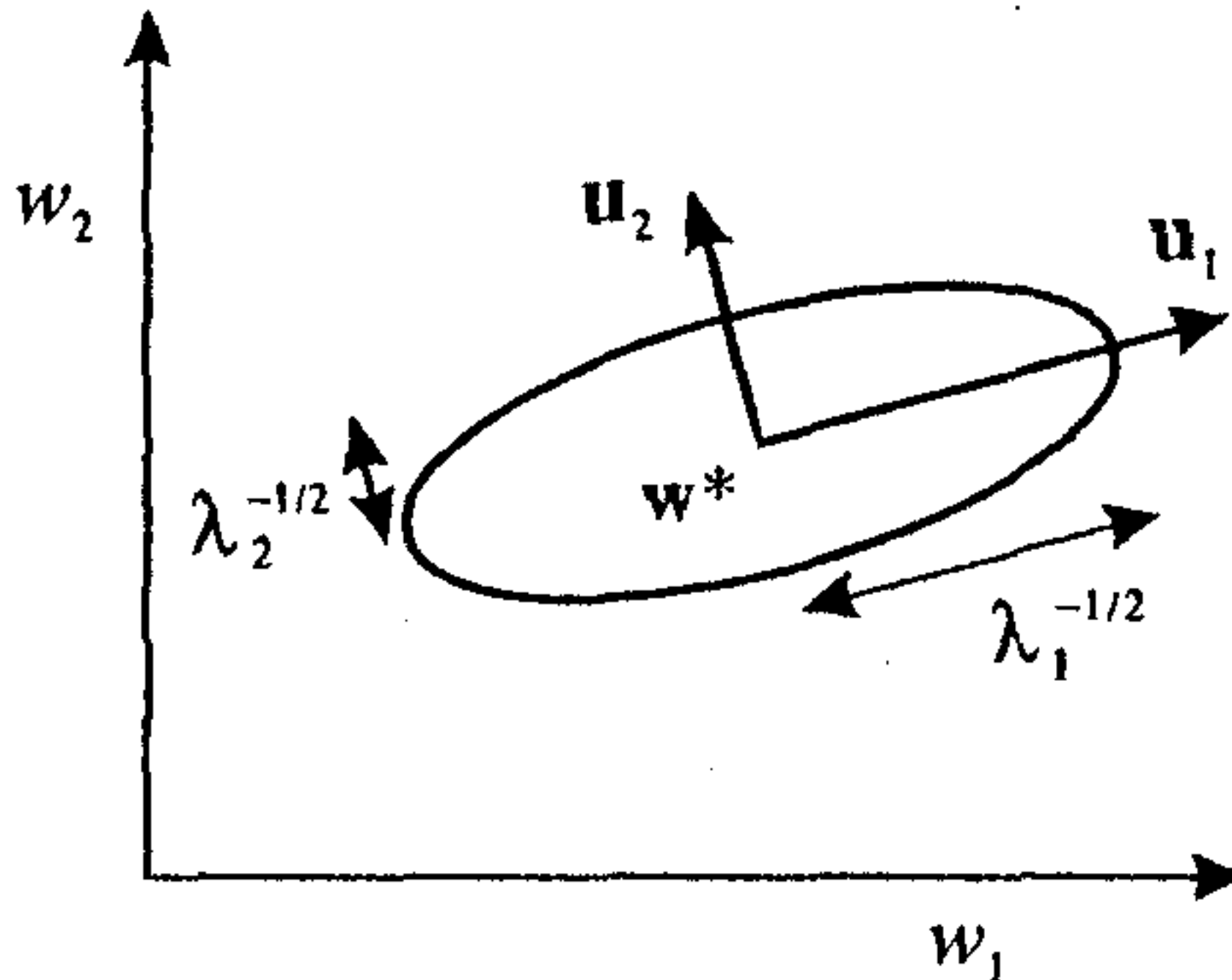
Saddle point



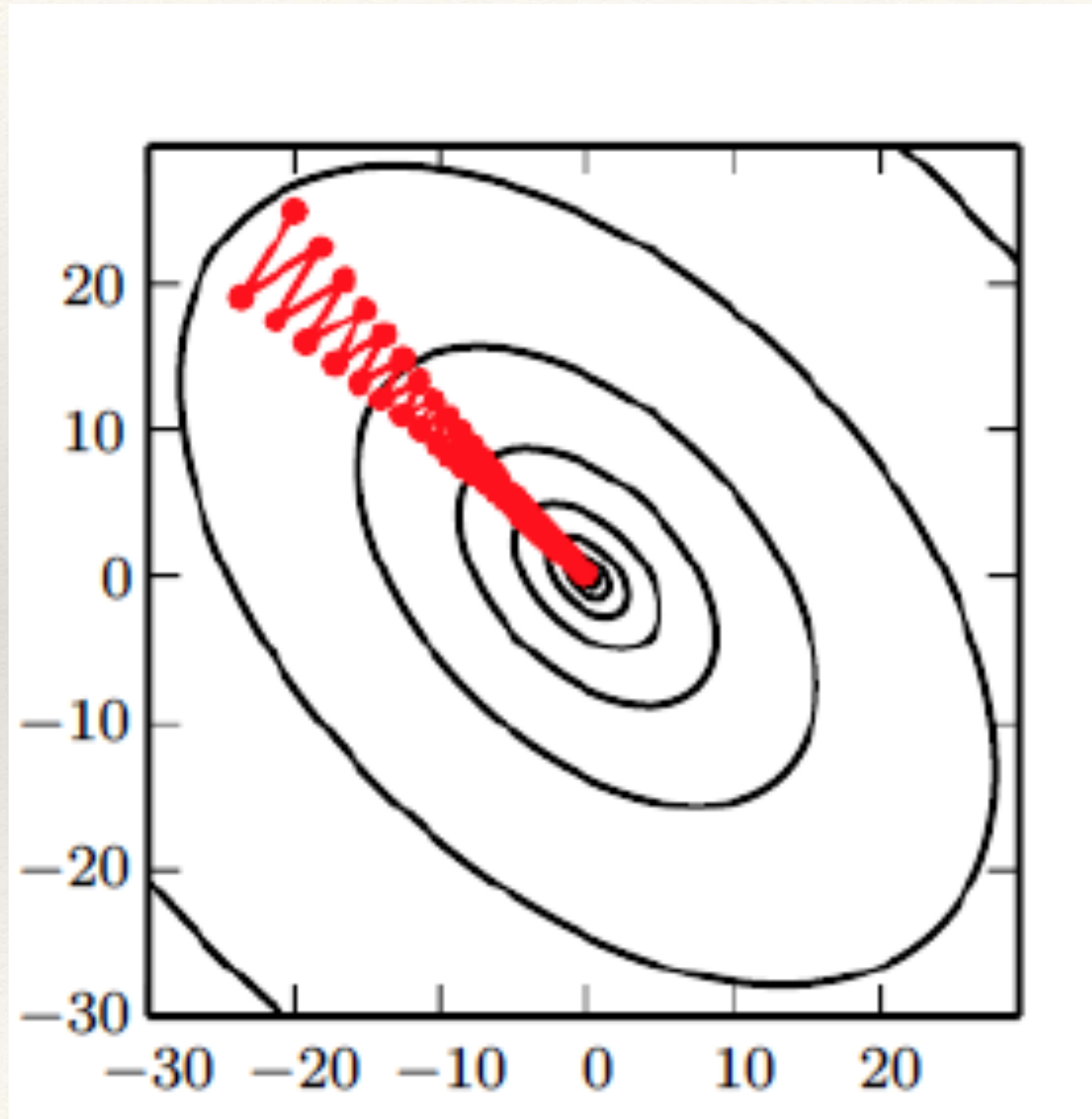
Gradient Descent Algorithm



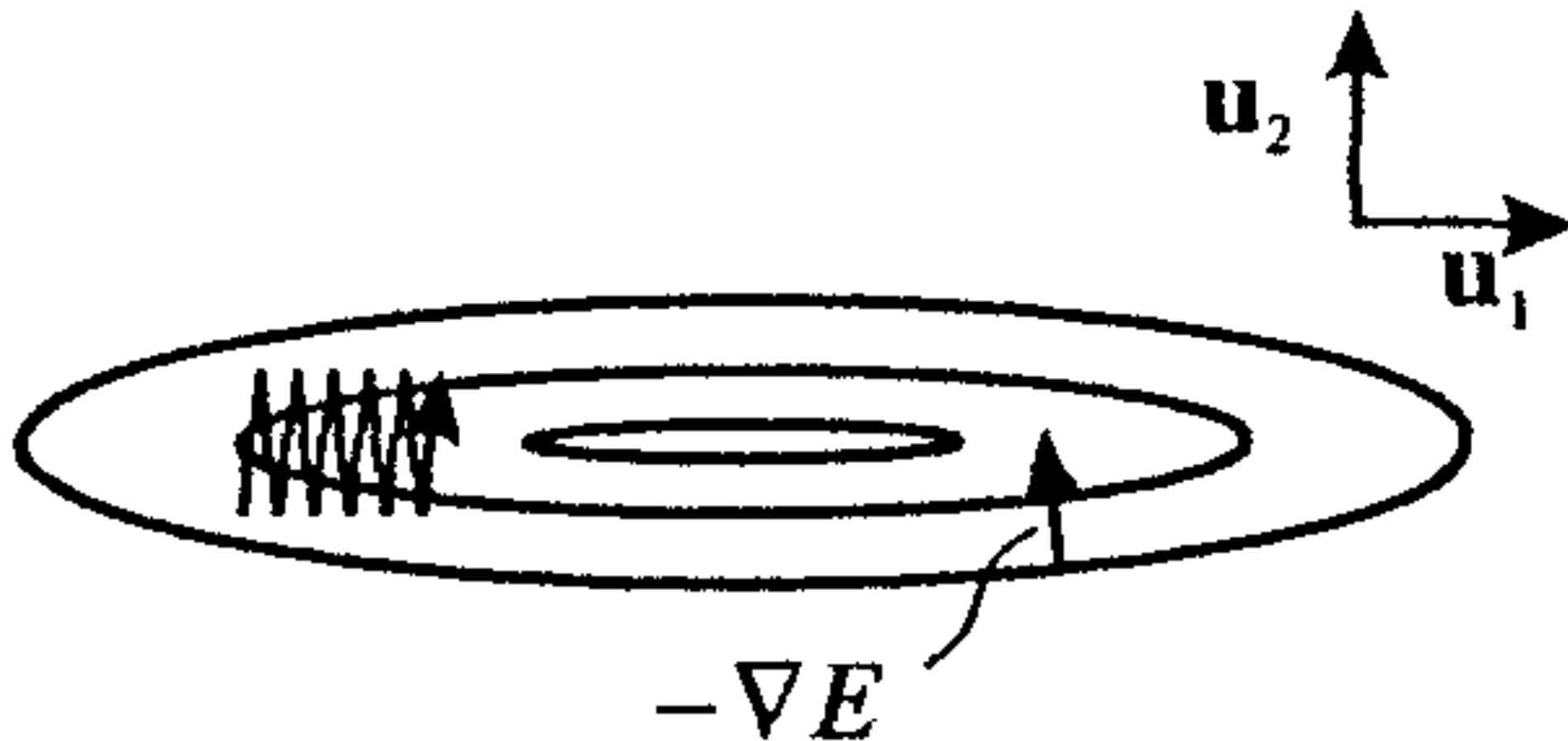
Weight Update Rule



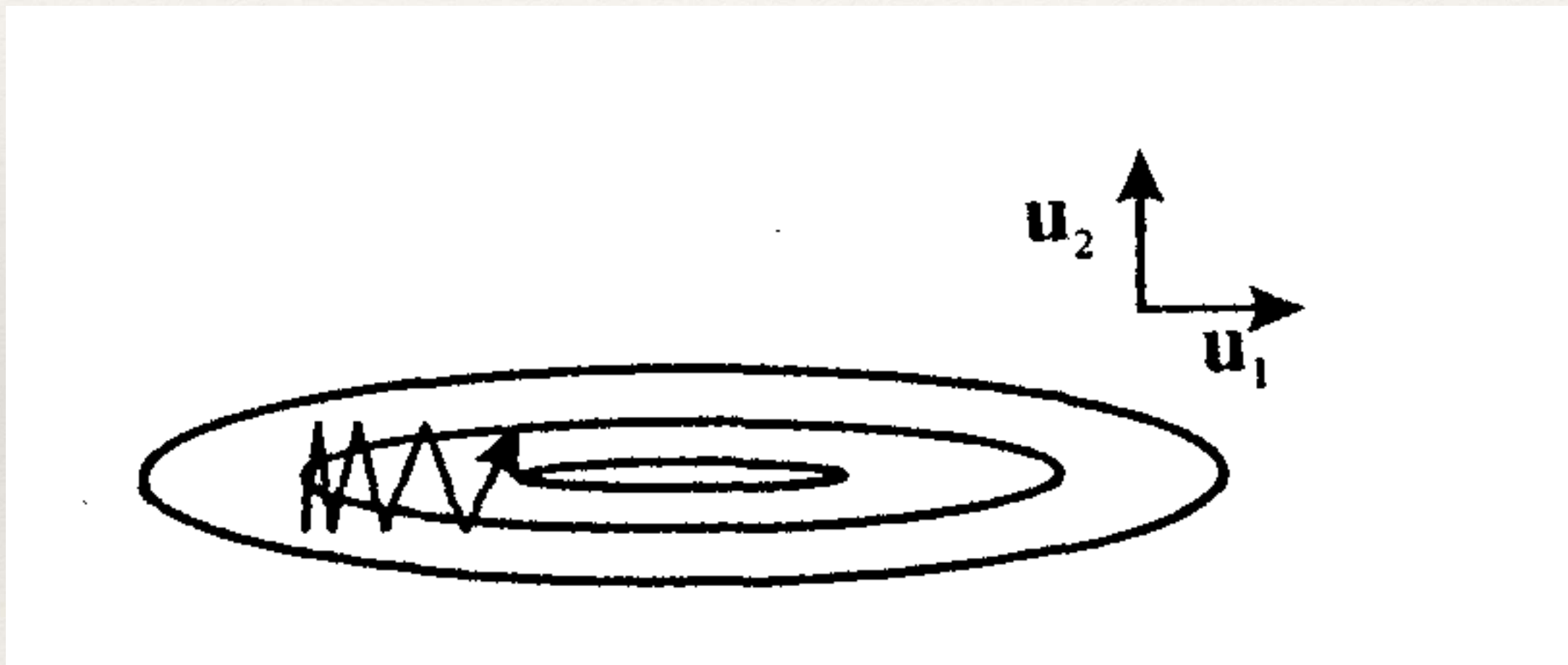
Slow Convergence of Gradient Descent



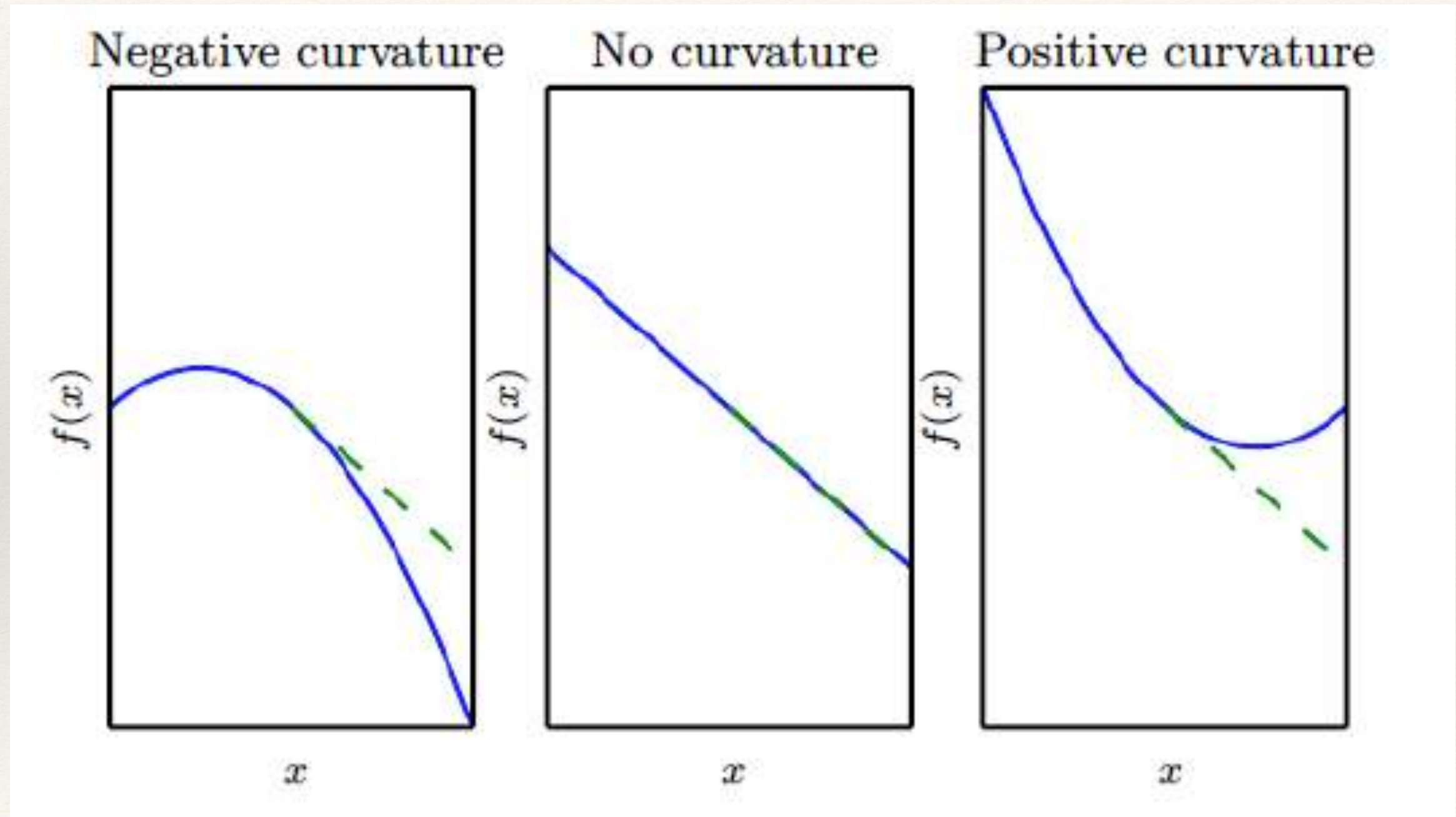
Gradient Descent Properties



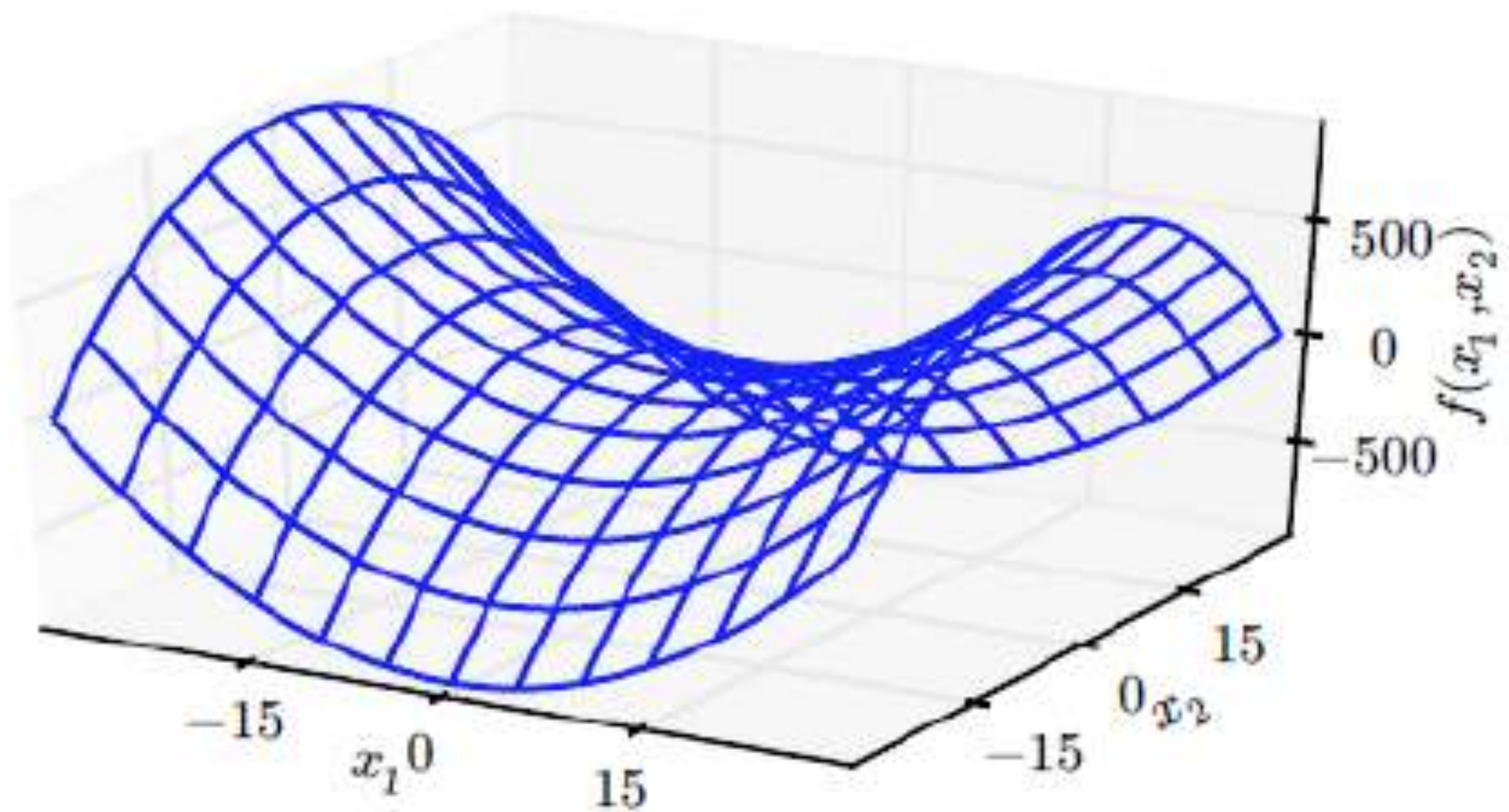
Gradient Descent With Momentum



Second Order Effects in Learning



Second Order Effects in Multi-dimensions



Newton's method

